

Chapitre 4

Solutions to the exercises

4.1 Solutions to the exercises from Chapter 3

4.1.1 Exercise 1

1. We have paired data $(U_i, V_i)_{1 \leq i \leq 9}$ where U_i and V_i correspond to the same i -th individual (the i -th movie) and represent, respectively, the age of the male lead actor and the age of the female lead actress in that movie.

We have access to the values of the differences $X_i = U_i - V_i$ and not only to their sign : we therefore prefer to use all the information available and use the Wilcoxon signed-rank test rather than the sign test.

Remark : we do not deal with checking the common-median assumption, nor the symmetry about that median, which are assumptions required for the Wilcoxon signed-rank test ; we simply assume they are satisfied.

2. Recall the assumptions of the Wilcoxon signed-rank test :
 - $X_1, \dots, X_n$ are independent ;
 - $X_1, \dots, X_n$ have continuous distributions ;
 - $X_1, \dots, X_n$ have a common median m ;
 - For $1 \leq i \leq n$, the distribution of X_i is symmetric about m .

The null hypothesis is $H_0 : m = 0$, and the alternative hypothesis is $H_1 : m > 0$. We use the test statistic

$$W_9^+ = \sum_{i=1}^9 R_{|X|}(i) \mathbf{1}_{\{X_i > 0\}}$$

where $(|X|_{(1)}, \dots, |X|_{(9)}) = (|X|_{\hat{\sigma}_{|X|}(1)}, \dots, |X|_{\hat{\sigma}_{|X|}(9)})$ and $R_{|X|} = \hat{\sigma}_{|X|}^{-1}$.

From the course, we know that under H_0 , W_9^+ has the same distribution as

$$\tilde{h}_9 := \sum_{j=1}^9 jY_j$$

where $Y_1, \dots, Y_9 \stackrel{iid}{\sim} Be(\frac{1}{2})$, and thus $\mathbb{E}_{H_0}[W_9^+] = \frac{9 \times (9+1)}{4} = 22.5$.

3. The observed values of $U_i, V_i, X_i, \mathbf{1}_{\{X_i > 0\}}, \hat{\sigma}_{|X|}(i), R_{|X|}(i)$ are :

i	1	2	3	4	5	6	7	8	9
u_i	40	57	44	52	39.2	77	48	36	39
v_i	39.5	55	41.5	49	39.5	84	79	30	23
x_i	0.5	2	2.5	3	-0.3	-7	-31	6	16
$\mathbf{1}_{\{x_i > 0\}}$	1	1	1	1	0	0	0	1	1
$\hat{\sigma}_{ x }(i)$	5	1	2	3	4	8	6	9	7
$R_{ x }(i)$	2	3	4	5	1	7	9	6	8

The observed value of W_9^+ is therefore

$$w_9^+ = R_{|x|}(1) + R_{|x|}(2) + R_{|x|}(3) + R_{|x|}(4) + R_{|x|}(8) + R_{|x|}(9) = 2 + 3 + 4 + 5 + 6 + 8 = 28.$$

We use the test $\varphi_\alpha(X_1^9) = \mathbb{1}_{\{W_9^+ > q_{1-\alpha}^{\mathcal{L}(\tilde{h}_9)}\}}$ where $q_{1-\alpha}^{\mathcal{L}(\tilde{h}_9)}$ is the $(1-\alpha)$ -quantile of the distribution of $\tilde{h}_9$. Since W_9^+ takes values in $\mathbb{N}$, we also have $\varphi_\alpha(X_1^9) = \mathbb{1}_{\{W_9^+ \geq q_{1-\alpha}^{\mathcal{L}(\tilde{h}_9)} + 1\}}$.

Computation of the observed p -value :

Let $w = w_9^+$ for simplicity and let F denote the CDF of $\tilde{h}_9$. Since W_9^+ is integer-valued, using properties of the quantile function,

$$\begin{aligned} p(x_1^n) &= \inf \left\{ \alpha \in]0, 1[: w > F^{(-1)}(1-\alpha) \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: w \geq F^{(-1)}(1-\alpha) + 1 \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: F(w-1) \geq 1-\alpha \right\} \\ &= 1 - F(w-1) \\ &= 1 - P(\tilde{h}_n \leq w-1) = P(\tilde{h}_n \geq w). \end{aligned}$$

This yields here :

$$p(x_1^9) = 1 - P(\tilde{h}_9 \leq 28-1) = 1 - 0.7148 = 0.2852.$$

4. We have $p(x_1^9) = 0.2852 > \alpha = 0.05$, so we do not reject H_0 at level $\alpha = 5\%$.

4.1.2 Exercise 2

Let $U_1, \dots, U_9$ be the numbers of adult worms found in the intestines of the 9 untreated mice, and $V_1, \dots, V_7$ the numbers of adult worms found in the intestines of the 7 treated mice.

Recall the assumptions of the Mann-Whitney test :

- $U_1, \dots, U_n$ are i.i.d. with the same distribution as U ;
- $V_1, \dots, V_p$ are i.i.d. with the same distribution as V ;
- U_1^n is independent of V_1^p ;
- U and V are continuous.

Remarks : the last assumption causes no difficulty here since there are no ties in the two series of values given in the statement.

Let F (resp. G) denote the CDF of U (resp. V).

We test

$$H_0 : F = G \text{ (i.e. the } U_i \text{ and the } V_j \text{ have the same distribution)}$$

against

$$H_1 : \text{the treatment is effective : the } V_j \text{ tend to be smaller than the } U_i.$$

Here $n = 9, p = 7$. Let $\Sigma_1 = \sum_{i=1}^9 R_i, \Sigma_2 = \sum_{j=1}^7 S_j$, where R_i (resp. S_j) is the rank of U_i (resp. V_j) in the pooled vector $(U_1, \dots, U_9, V_1, \dots, V_7)$, Z the ordered pooled vector, $M_U = \Sigma_1 - \frac{n \times (n+1)}{2} = \Sigma_1 - 45$ and $M_V = \Sigma_2 - \frac{p \times (p+1)}{2} = \Sigma_2 - 28$. We thus have $M_V = np - M_U = 63 - M_U$.

We know, from Proposition 3.38, that under H_0 , M_U and M_V have the same distribution as $\tilde{h}_{9,7}$ with CDF $F_{9,7}$, and are free of F .

We may use a test of the form $\varphi_\alpha(U_1^9, V_1^7) = \mathbb{1}_{\{M_V < k_\alpha\}}$ or a test of the form $\varphi'_\alpha(U_1^9, V_1^7) = \mathbb{1}_{\{M_U > k'_\alpha\}}$: we expect that under H_1 , M_V is small and M_U is large.

We have $P_{H_0}(M_U > k'_\alpha) = 1 - P_{H_0}(M_U \leq k'_\alpha) = 1 - F_{9,7}(k'_\alpha)$ so $P_{H_0}(M_U > k'_\alpha) \leq \alpha \Leftrightarrow 1 - \alpha \leq F_{9,7}(k'_\alpha)$, and we therefore take $k'_\alpha = q_{1-\alpha}^{\mathcal{L}(\tilde{h}_{9,7})}$, i.e. the $(1 - \alpha)$ -quantile of the distribution of $\tilde{h}_{9,7}$. Since $M_V = 63 - M_U$, we have $\mathbb{P}_{H_0}(M_V < k_\alpha) = \mathbb{P}_{H_0}(63 - k_\alpha < M_U)$ and thus $\mathbb{P}_{H_0}(M_V < k_\alpha) \leq \alpha \Leftrightarrow 1 - \alpha \leq F_{9,7}(63 - k_\alpha)$, so we take $k_\alpha = 63 - q_{1-\alpha}^{\mathcal{L}(\tilde{h}_{9,7})}$.

Since M_U and M_V take values in $\mathbb{N}$, this amounts to using either of the two equivalent tests :

$$\begin{aligned}\varphi_\alpha(U_1^9, V_1^7) &= \mathbb{1}_{\{M_V \leq 62 - q_{1-\alpha}^{\mathcal{L}(\tilde{h}_{9,7})}\}} \\ \varphi'_\alpha(U_1^9, V_1^7) &= \mathbb{1}_{\{M_U \geq q_{1-\alpha}^{\mathcal{L}(\tilde{h}_{9,7})} + 1\}}.\end{aligned}$$

Let $\sigma_1, \sigma_2, r_i, s_j, u_i, v_j, z_k, m_u, m_v$ denote the observed values of the variables $\Sigma_1, \Sigma_2, R_i, S_j, Z_k, M_U, M_V$. We observe :

k	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
z_k	47	49	51	53	55	57	60	61	62	63	65	67	68	71	75	79
Index in u	-	-	1	-	2	-	-	-	3	4	5	-	6	7	8	9
Index in v	1	2	-	3	-	4	5	6	-	-	-	7	-	-	-	-

Hence :

i	1	2	3	4	5	6	7	8	9
r_i	3	5	9	10	11	13	14	15	16

so $\sigma_1 = 3 + 5 + 9 + 10 + 11 + 13 + 14 + 15 + 16 = 96$, $m_u = \sigma_1 - 45 = 51$, and

j	1	2	3	4	5	6	7
s_j	1	2	4	6	7	8	12

so $\sigma_2 = 1 + 2 + 4 + 6 + 7 + 8 + 12 = 40$, $m_v = \sigma_2 - 28 = 12$.

Remark : from Proposition 3.38, we know that M_V corresponds to the number of pairs (U_i, V_j) such that $U_i < V_j$ among all possible pairs.

Here we observe

$$\begin{aligned}\{(i, j); u_i < v_j\} &= \\ &= \{(1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (2, 4), (2, 5), (2, 6), (2, 7), (3, 7), (4, 7), (5, 7)\}\end{aligned}$$

and we indeed recover $m_v = 12$.

Computation of the observed p -value : let $F_{9,7}$ denote the CDF of $\tilde{h}_{9,7}$.

$$\begin{aligned}p(u_1^9, v_1^7) &= \inf \left\{ \alpha \in]0, 1[: m_u > F_{9,7}^{(-1)}(1 - \alpha) \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: m_u \geq F_{9,7}^{(-1)}(1 - \alpha) + 1 \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: F_{9,7}(m_u - 1) \geq 1 - \alpha \right\} \\ &= 1 - F_{9,7}(m_u - 1) \\ &= 1 - P(\tilde{h}_{9,7} \leq m_u - 1) = P(\tilde{h}_{9,7} \geq m_u).\end{aligned}$$

This yields here :

$$p(u_1^9, v_1^7) = 1 - F_{9,7}(50).$$

However, the statement does not give us this quantity. We can proceed by using the symmetry of the distribution of $\tilde{h}_{9,7}$ around $\frac{np}{2} = 63/2$:

$$1 - F_{9,7}(50) = P(\tilde{h}_{9,7} \geq 51) = P(2 \times 63/2 - \tilde{h}_{9,7} \geq 51) = P(63 - \tilde{h}_{9,7} \geq 51) = F_{9,7}(12) = 0.021.$$

For a risk level $\alpha < p(u_1^9, v_1^7) = 0.021$, we do not reject H_0 at level α (we cannot conclude that the treatment is effective) ;

For a risk level $\alpha > p(u_1^9, v_1^7) = 0.021$, we reject H_0 at level α (at risk level α of being wrong, we believe the treatment is effective).

4.1.3 Exercise 3

Let U_i (resp. V_i) denote the effectiveness of the new treatment (A) (resp. the old treatment (B)) for individual i , for $1 \leq i \leq n$, with $n = 18$. Let $X_i = U_i - V_i$. We only have access to the sign of the X_i , i.e. to $Y_i := \mathbb{1}_{\{X_i > 0\}}$, since patients are only asked to say whether treatment (A) was more effective than treatment (B) (in which case $X_i > 0$) or whether treatment (A) was less effective than treatment (B) (in which case $X_i < 0$).

There is no possible zero answer, which corresponds to $\mathbb{P}(X_i = 0) = 0$. We will therefore use the sign test. Recall the assumptions required for this test :

- $X_1, \dots, X_n$ are independent ;
- $X_1, \dots, X_n$ have a common median m ;
- For $1 \leq i \leq n$, $\mathbb{P}(X_i = m) = 0$.

We test

$$H_0 : m = 0 \text{ (which models "the two treatments have the same effectiveness")}$$

against

$$H_1 : m > 0 \text{ (which models "the new treatment is better than the old one")}$$

Under H_0 , $Y_1, \dots, Y_{18} \stackrel{iid}{\sim} Be(\frac{1}{2})$ because, since under H_0 the median is 0, we have $\mathbb{P}(X_i \geq 0) \geq \frac{1}{2}$ and $\mathbb{P}(X_i \leq 0) \geq \frac{1}{2}$, but since moreover $\mathbb{P}(X_i = 0) = 0$, $\mathbb{P}(Y_i = 1) = \mathbb{P}(X_i > 0) = \frac{1}{2}$. We use the test

$$\varphi_\alpha(X_1^{18}) = \mathbb{1}_{\{\sum_{i=1}^{18} \mathbb{1}_{\{X_i > 0\}} > q_{1-\alpha}^{B(18, \frac{1}{2})}\}}$$

or, since $\sum_{i=1}^{18} \mathbb{1}_{\{X_i > 0\}}$ is integer-valued,

$$\varphi_\alpha(X_1^{18}) = \mathbb{1}_{\{\sum_{i=1}^{18} \mathbb{1}_{\{X_i > 0\}} \geq q_{1-\alpha}^{B(18, \frac{1}{2})} + 1\}}.$$

Let x_i denote the observed value of X_i . The observed value of $\sum_{i=1}^{18} \mathbb{1}_{\{X_i > 0\}}$ is therefore $\sum_{i=1}^{18} \mathbb{1}_{\{x_i > 0\}} = 12$.

Computation of the observed p -value : let F denote the CDF of $B(18, 1/2)$.

$$\begin{aligned} p(x_1^{18}) &= \inf \left\{ \alpha \in]0, 1[: 12 > F^{(-1)}(1 - \alpha) \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: 12 \geq F^{(-1)}(1 - \alpha) + 1 \right\} \\ &= \inf \left\{ \alpha \in]0, 1[: F(12 - 1) \geq 1 - \alpha \right\} \\ &= 1 - F(11). \end{aligned}$$

This yields here :

$$p(x_1^{18}) = 1 - F(11) \approx 0.23.$$

For a risk level $\alpha < p(x_1^{18}) = 0.23$, we do not reject H_0 at level α (we cannot conclude that the new treatment is more effective than the old one) ;

For a risk level $\alpha > p(x_1^{18}) = 0.23$, we reject H_0 at level α (at risk level α of being wrong, we believe the new treatment is more effective than the old one).

4.1.4 Proof of Proposition 3.37

1. It is immediate to see that $\Sigma_1 \geq 1 + 2 + \dots + n = \frac{n(n+1)}{2}$ and $\Sigma_2 \geq 1 + 2 + \dots + p = \frac{p(p+1)}{2}$. Since $\Sigma_1 + \Sigma_2 = \sum_{i=1}^{n+p} i = \frac{(n+p)(n+p+1)}{2}$, we have

$$\Sigma_1 \leq \frac{(n+p)(n+p+1)}{2} - \frac{p(p+1)}{2} = np + \frac{n(n+1)}{2}.$$

Similarly, $\Sigma_2 \leq np + \frac{p(p+1)}{2}$.

2. We now work under H_0 .

- (a) Then all variables $U_1, \dots, U_n, V_1, \dots, V_p$ are i.i.d. Hence we have a pooled i.i.d. sample of size $N = n + p$. By Proposition 3.31(case $s = 1$), for every i , the r.v. R_i , which is a component of the rank vector of the pooled sample $R_{U,V}$, follows a uniform distribution on $\{1, \dots, N\}$. The same holds for each S_j . Therefore, the expectation and variance of each of these variables are simply the expectation and variance of a uniform random variable on $\{1, \dots, N\}$. Thus, for all $i = 1, \dots, n$ and for all $j = 1, \dots, p$,

$$\mathbb{E}(R_i) = \mathbb{E}(S_j) = \frac{1}{N} \sum_{i=1}^N i = \frac{N+1}{2} = \frac{n+p+1}{2}$$

$$\begin{aligned} \mathbf{Var}(R_i) &= \mathbf{Var}(S_j) = \frac{1}{N} \sum_{i=1}^N i^2 - \left(\frac{N+1}{2}\right)^2 = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2 \\ &= \frac{N^2 - 1}{12}. \end{aligned}$$

- (b) Therefore,

$$\begin{aligned} \mathbb{E}\Sigma_1 &= \mathbb{E}(R_1 + \dots + R_n) = n\mathbb{E}R_1 = \frac{n(n+p+1)}{2} \\ \mathbb{E}\Sigma_2 &= \mathbb{E}(S_1 + \dots + S_p) = p\mathbb{E}S_1 = \frac{p(n+p+1)}{2}. \end{aligned}$$

- (c) It remains to compute the variances of Σ_1 and Σ_2 . Note that the variables R_i and S_j have the same distribution but are not independent! We have

$$\mathbf{Var}\Sigma_1 = \mathbf{Var}(R_1 + \dots + R_n) = \sum_{i=1}^n \mathbf{Var}(R_i) + \sum_{i=1}^n \sum_{j \neq i}^n \mathbf{Cov}(R_i, R_j).$$

We have already computed the variances, so we must now compute the covariances. Let $i \neq j$,

$$\begin{aligned} \mathbf{Cov}(R_i, R_j) &= \mathbb{E}\left[(R_i - \mathbb{E}R_i)(R_j - \mathbb{E}R_j)\right] \\ &= \sum_{1 \leq k, l \leq N, k \neq l} \left(k - \frac{N+1}{2}\right)\left(l - \frac{N+1}{2}\right) \mathbb{P}(R_i = k, R_j = l). \end{aligned}$$

Now (R_i, R_j) has the same distribution as (R_1, R_2) and, for $k \neq l$,

$$\mathbb{P}(R_1 = k, R_2 = l) = \frac{1}{N(N-1)}.$$

Thus

$$\mathbf{Cov}(R_i, R_j) = \mathbf{Cov}(R_1, R_2) = \frac{1}{N(N-1)} \sum_{k \neq l} \left(k - \frac{N+1}{2}\right)\left(l - \frac{N+1}{2}\right).$$

But

$$\begin{aligned}\sum_{k \neq l} (k - \frac{N+1}{2})(l - \frac{N+1}{2}) &= \sum_{1 \leq k, l \leq N} (k - \frac{N+1}{2})(l - \frac{N+1}{2}) - \sum_{k=1}^N (k - \frac{N+1}{2})^2 \\ &= \left[\sum_{k=1}^N (k - \frac{N+1}{2}) \right]^2 - \sum_{k=1}^N (k - \frac{N+1}{2})^2.\end{aligned}$$

Moreover,

$$\sum_{k=1}^N (k - \frac{N+1}{2}) = \sum_{k=1}^N k - \frac{N(N+1)}{2} = 0.$$

Hence

$$\mathbf{Cov}(R_i, R_j) = -\frac{1}{N(N-1)} \sum_{k=1}^N (k - \frac{N+1}{2})^2 = -\frac{1}{N-1} \mathbf{Var} R_1.$$

Finally,

$$\begin{aligned}\mathbf{Var}(\Sigma_1) &= \sum_{i=1}^n \mathbf{Var}(R_i) + \sum_{i=1}^n \sum_{j \neq i} \mathbf{Cov}(R_i, R_j) \\ &= n \mathbf{Var}(R_1) + n(n-1) \mathbf{Cov}(R_1, R_2) \\ &= n \mathbf{Var}(R_1) - \frac{n(n-1)}{N-1} \mathbf{Var}(R_1) \\ &= \frac{n(N-n)}{N-1} \frac{N^2-1}{12} \\ &= \frac{n(N-n)(N+1)}{12} \\ &= \frac{np(n+p+1)}{12}.\end{aligned}$$

The computation of $\mathbf{Var}(\Sigma_2)$ follows from that of $\mathbf{Var}(\Sigma_1)$ by exchanging the roles of n and p .